

A Counterexample to Compactness Without the Zero- Δ_2 Condition

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Source and Target

The source paper is Pierre-A. Vuillermot, *On the compactness of embeddings for a class of weighted Orlicz-Sobolev sequence spaces*, arXiv:2503.19454. The target is the remark after Theorem 4 on PDF page 17, source lines 1188–1195. In the notation of the source paper, the author asks whether the compact embedding

$$\mathfrak{s}_{k',\Phi,w} \hookrightarrow \mathfrak{s}_{k,\Phi,w}, \quad k' > k,$$

still holds when $\mathfrak{s}_{k,\Phi,w}$ is strictly included in $l_{k,\Phi,w}$, and also whether compact embeddings

$$l_{k',\Phi,w} \hookrightarrow l_{k,\Phi,w}, \quad k' > k,$$

exist in that situation.

proof is (32), which follows from the Δ_2 -condition at zero thanks to (22). But that condition also implies (33), so that the question whether the compact embedding (34) still holds whenever $\mathfrak{s}_{k,\Phi,w}$ is strictly included in $l_{k,\Phi,w}$ remains open at this time. By the same token, it is not known whether there exist compact embeddings of the form $l_{k',\Phi,w} \hookrightarrow l_{k,\Phi,w}$ whenever $k' > k$.

Finally, there are many ways to combine the preceding results to generate new compact embeddings. For instance, the following theorem is an immediate

17

Proof Intuition

The source proof of compactness uses the zero- Δ_2 condition to make tails uniformly small after passing from the k' -weight to the k -weight. If Φ is very flat at the origin, this uniform tail estimate can fail exactly on one-coordinate vectors. Choosing $\Phi(t) = e^{-1/t}$ near zero makes the inverse behave like $1/\log(1/u)$. For the far-out coordinate n , the k' -unit vector has size

$$\frac{1}{k' \log(1 + \Phi(n))},$$

and its k -norm is therefore the constant k/k' , independent of n . These vectors cannot have a compact image.

Counterexample

Theorem 1. *For every pair $0 < k < k'$, there are an Orlicz function Φ and weights $w_m \equiv 1$ such that:*

1. $\mathfrak{s}_{k,\Phi,w} \subsetneq l_{k,\Phi,w}$;
2. the embedding $\mathfrak{s}_{k',\Phi,w} \rightarrow \mathfrak{s}_{k,\Phi,w}$ is continuous but not compact;
3. the embedding $l_{k',\Phi,w} \rightarrow l_{k,\Phi,w}$ is continuous but not compact.

Consequently the compactness questions in the source remark have a negative answer in general.

Proof. Fix $0 < k < k'$. Let $0 < a < 1/2$, for instance $a = 1/4$, and define

$$\Phi(0) = 0, \quad \Phi(t) = e^{-1/t} \quad (0 < t \leq a),$$

with a linear extension for $t \geq a$:

$$\Phi(t) = \Phi(a) + \Phi'(a)(t - a), \quad t \geq a.$$

Since $e^{-1/t}$ has second derivative

$$e^{-1/t} \frac{1 - 2t}{t^4}$$

on $(0, a]$, this function is convex on $[0, a]$. The linear extension uses the tangent slope at a , so the full function is continuous, convex, increasing, positive on $(0, \infty)$, and unbounded. Thus it is an Orlicz function in the sense of the source paper. It fails the zero- Δ_2 condition because

$$\frac{\Phi(2t)}{\Phi(t)} = \exp\left(\frac{1}{2t}\right) \rightarrow \infty \quad (t \downarrow 0).$$

Set $w_m = 1$ for all $m \in \mathbb{Z}$.

We first show that $\mathfrak{s}_{k,\Phi,w}$ is strictly smaller than $l_{k,\Phi,w}$. For all sufficiently large positive integers m , set

$$p_m = \frac{1}{(k+2)\log(1+\Phi(m))},$$

and set $p_m = 0$ for the remaining indices. Since the extension of Φ is linear at infinity, $1 + \Phi(m)$ is comparable to m . Also $p_m < a$ for large m . Hence

$$\Phi(p_m) = (1 + \Phi(m))^{-(k+2)}$$

for large m , and therefore

$$\sum_m (1 + \Phi(|m|))^k \Phi(|p_m|) < \infty.$$

Thus $p = (p_m) \in l_{k,\Phi,w}$. On the other hand, take $\rho = (k+1)/(k+2)$. Then, for large m ,

$$\Phi(p_m/\rho) = (1 + \Phi(m))^{-(k+1)},$$

so

$$\sum_m (1 + \Phi(|m|))^k \Phi(|p_m|/\rho) \geq \sum_{m \gg 1} (1 + \Phi(m))^{-1} = \infty,$$

again because $\Phi(m)$ is linear at infinity. Thus $p \notin \mathfrak{s}_{k,\Phi,w}$.

Now define, for large positive integers n ,

$$x^{(n)} = \frac{1}{k' \log(1 + \Phi(n))} e_n,$$

where e_n is the n -th coordinate vector. These vectors have finite support, hence belong to every small space $\mathfrak{s}_{j,\Phi,w}$. Put $U_n = 1 + \Phi(n)$. Since the nonzero coordinate is in the $e^{-1/t}$ -range of Φ , we have

$$\Phi\left(\frac{1}{k' \log U_n}\right) = U_n^{-k'}.$$

The Luxemburg norm of a one-coordinate vector be_n in $l_{j,\Phi,w}$ is

$$\|be_n\|_{j,\Phi,w} = \frac{|b|}{\Phi^{-1}((1 + \Phi(n))^{-j})}.$$

Consequently

$$\|x^{(n)}\|_{k',\Phi,w} = 1, \quad \|x^{(n)}\|_{k,\Phi,w} = \frac{k}{k'}.$$

Thus $(x^{(n)})$ is contained in the unit ball of $\mathfrak{s}_{k',\Phi,w}$ and of $l_{k',\Phi,w}$, but its image in the k -space has constant norm k/k' .

For each coordinate n , the coordinate projection P_n is contractive for the Luxemburg norm, since deleting coordinates can only decrease the defining modular. Hence, if $n \neq m$,

$$\|x^{(n)} - x^{(m)}\|_{k,\Phi,w} \geq \|P_n(x^{(n)} - x^{(m)})\|_{k,\Phi,w} = \|x^{(n)}\|_{k,\Phi,w} = \frac{k}{k'}.$$

The image sequence has no Cauchy subsequence in $l_{k,\Phi,w}$, and a fortiori no Cauchy subsequence in the closed subspace $\mathfrak{s}_{k,\Phi,w}$. Therefore neither embedding is compact. The continuity of both embeddings for $k' > k$ is the elementary continuous embedding already recorded in the source paper before Theorem 3. $\square$

Verification Notes

No numerical computation is used. The key checks are:

- the function Φ is a valid Orlicz function and fails the zero- Δ_2 condition;
- the sequence $p_m = [(k+2) \log(1 + \Phi(m))]^{-1}$ proves strict inclusion $\mathfrak{s}_{k,\Phi,w} \subsetneq l_{k,\Phi,w}$;
- the one-coordinate norm formula gives the constant image norm k/k' ;
- contractive coordinate projections prevent any Cauchy subsequence.

Novelty Check

The run indexes were searched for arXiv:2503.19454, the source title, compact embeddings, and the exact notation $\mathfrak{s}_{k,\Phi,w}$, $l_{k,\Phi,w}$; no duplicate packet or prior attempt was found. A bounded web search on 2026-07-03 for exact title/id phrases and the displayed compactness notation found the source arXiv page but no later answer to this exact question. This packet should therefore be reviewed as a new candidate counterexample, not as a literature-status identification.

References

- P.-A. Vuillermot, *On the compactness of embeddings for a class of weighted Orlicz-Sobolev sequence spaces*, arXiv:2503.19454, 2025.